Chapter 16, Problem 102P

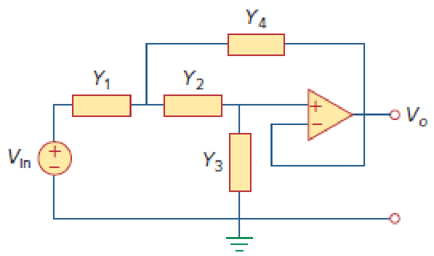
Problem

Synthesize the transfer function

$$\frac{V_o(s)}{V_{in}(s)} = \frac{10^6}{s^2 + 100s + 10^6}$$

using the topology of Fig. 16.111. Let $Y1 = 1/R1$, $Y2 = 1/R2$, $Y3 = sC1$, $Y4 = sC2$. Choose $R1$ = 1 kΩ and determine $C1$, $C2$, and $R2$.

Figure 16.111:

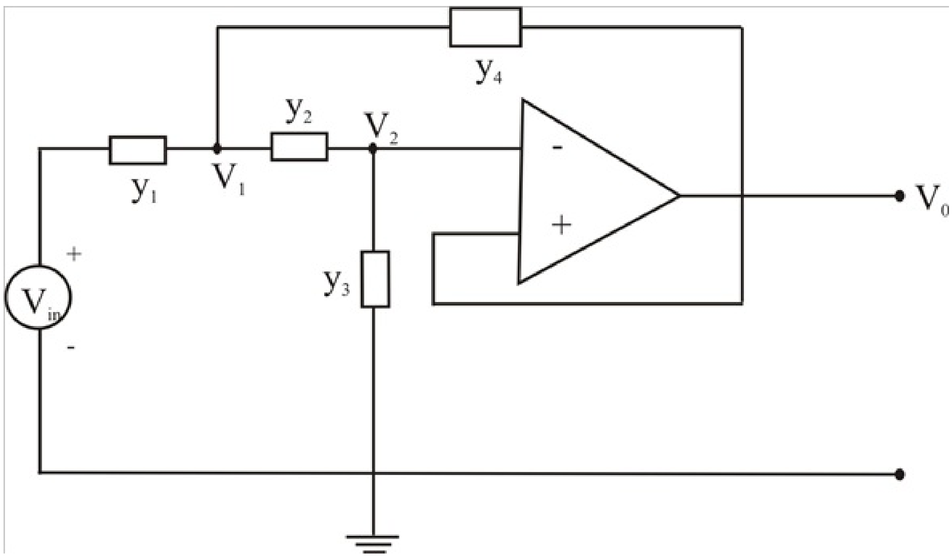


Step-by-step solution

Step 1 of 4

Consider the circuit shown below, we notice that

$$V_3 = V_0 \quad and \quad V_2 = V_3 = V_O$$



Step 2 of 4

At node 1,

$$(V_{in} - V_1)Y_1 = (V_1 - V_O)Y_2 + (V_1 - V_O)Y_4$$

$$V_{in}Y_1 = V_1(Y_1 + Y_2 + Y_4) - V_O(Y_2 + Y_4) \qquad -(1)$$

At node 2,

$$(V_1 - V_O)Y_2 = (V_O - 0)Y_3$$

$$V_1Y_2 = (Y_2 + Y_3)V_O$$

$$V_{12} = \frac{Y_2 + Y_3}{Y_2}V_O \qquad -(2)$$

Step 3 of 4

Subsituting (2)in to (1)

$$V_{in}Y_1 = \frac{Y_2+Y_3}{Y_2}(Y_1+Y_2+Y_4)V_O - V_O(Y_2+Y_4)$$

$$V_{in}Y_1Y_2 = V_O\left(Y_1Y_2+Y_2^2+Y_2Y_4+Y_1Y_3+Y_2Y_3+Y_3Y_4-Y_2^2-Y_2Y_4\right)$$

$$\frac{V_O}{V_{in}} = \frac{Y_1Y_2}{(Y_1Y_2+Y_1Y_3+Y_2Y_3+Y_3Y_4)}$$

Y_1 and Y_2 must be resistive , while Y_3 and Y_4 must be capacitive

let $\quad Y_1 = \frac{1}{R_1} \qquad Y_2 = \frac{1}{R_2} \qquad Y_3 = SC, \qquad Y_4 = SC_2$

$$\frac{V_O}{V_{in}} = \frac{\frac{1}{R_1R_2}}{\frac{1}{R_1R_2}+\frac{SC_1}{R_1}+\frac{SC_1}{R_2}+S^2C_1C_2}$$

Step 4 of 4

Choose $\quad R_1 = 1K\Omega \quad$ *then* $\quad \frac{1}{R_1R_2C_1C_2} = 10^6 \quad$ *and* $\quad \frac{R_1+R_2}{R_1R_2C_2} = 100$

We have three equations and four unknowns. Thus there is a fainily of solutions, one such solution is

$R_2 = 1K\Omega \qquad C_1 = 50\eta F \qquad C_2 = 20\mu F$